

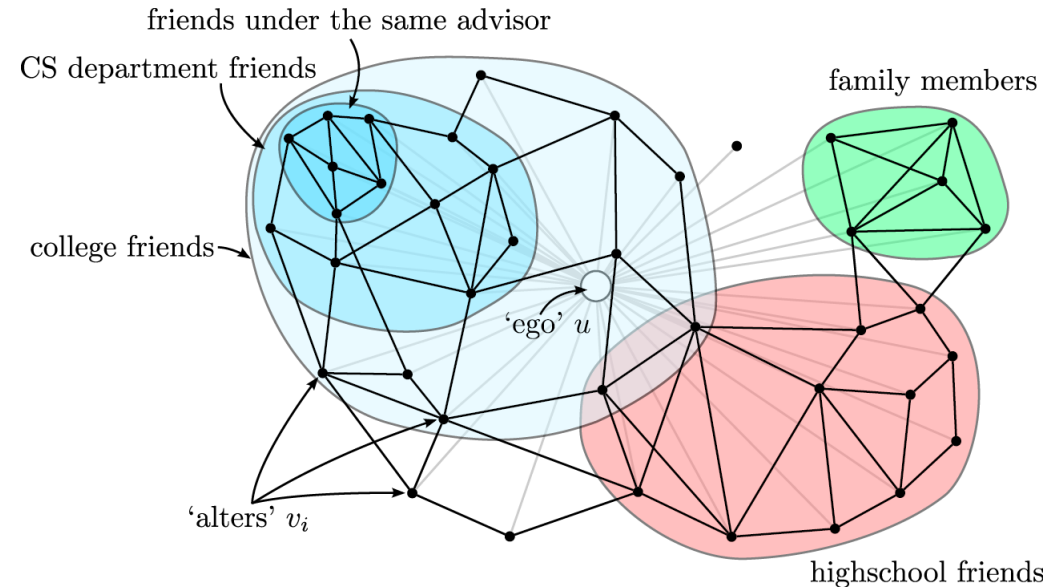
Mining Social Networks

COMPSCI 753

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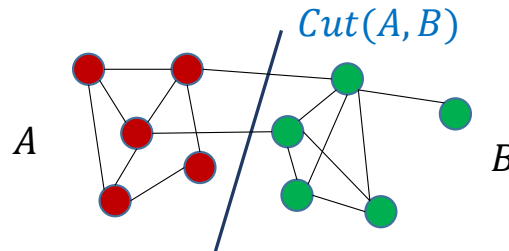
Community detection

- Girvan-Newman algorithm
- **Spectral clustering**
- Modularity maximization



Community detection using graph partitioning

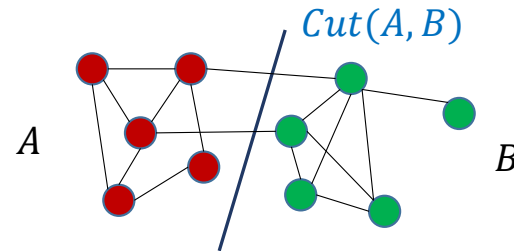
- Graph partitioning: find best cuts into communities
- Graph cut
 - Bi-partitioning - Consider partitioning the graph into two parts A and B



- A cut $Cut(A, B)$ partition the graph into two disjoint groups A, B
 - $Cut A, B$ is the sum of weights of edges that connect a node in A to a node in B
 - $Cut(A, B) = \sum_{i \in A, j \in B} w_{ij} = 1 + 1 = 2$ (assuming weight of each edge = 1)

What is a good partition?

- We want each community to be dense subgraph
 - Maximize the number of **within-group connections**
- We want to make each community has as less connections with other communities as possible
 - Minimize the number of **between-group connections**

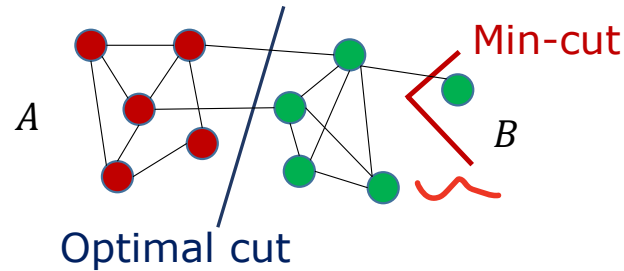


Graph partitioning criterion

- Criterion 1: **Min-cut**

- Minimize weight of connections between groups:

$$\arg \min_{A,B} \text{cut}(A, B)$$



inbalance:

- Problems:

- Only considers between – group connections
- Imbalance groups – communities may not necessarily be of the same size

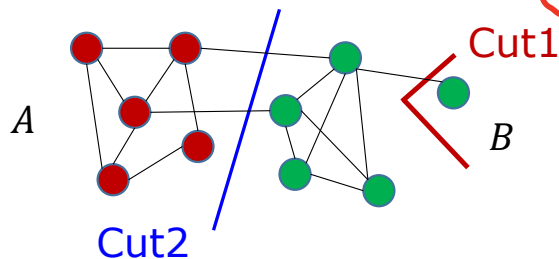
Graph partitioning criterion

■ Criterion 2: **RatioCut**

- Normalize the value by the **size** (number of nodes) of both A and B this will produce partitions that are more similar in terms of number of nodes
- **Minimize** weight of connections between groups:

$$\text{RatioCut}(A, B) = \frac{\text{cut}(A, B)}{|A|} + \frac{\text{cut}(A, B)}{|B|}$$

Handwritten notes: 切边数 (cut edge count) above cut(A, B); A组node数 (A group node count) below |A|.



$$\text{RatioCut}(A_{\text{cut1}}, B_{\text{cut1}}) = \frac{1}{9} + \frac{1}{1} \approx 1.11$$

$$\text{RatioCut}(A_{\text{cut2}}, B_{\text{cut2}}) = \frac{2}{5} + \frac{2}{5} = 0.8$$

cost of cut.



min score.

Graph partitioning criterion

- Criterion 3: **Normalized-cut (NCut)** [Shi-Malik, '97]
 - Normalize the value by the **volume** of both A and B
 - Minimize weight of connections between groups:

$$NCut(A, B) = \frac{cut(A, B)}{vol(A)} + \frac{cut(A, B)}{vol(B)}$$
+ 几条边

$NCut(A_{cut1}, B_{cut1}) = \frac{1}{31} + \frac{1}{1} \approx 1.032$

$NCut(A_{cut2}, B_{cut2}) = \frac{2}{16} + \frac{2}{16} = 0.25$

✓

Degree of a node:

$$d_i = \sum_j w_{ij}$$

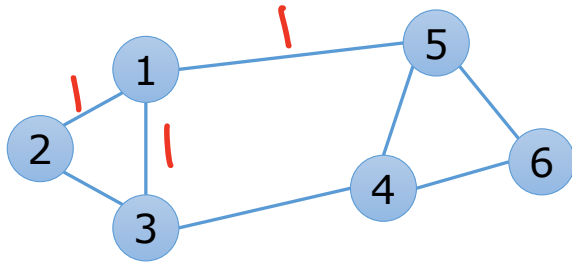
Volume of a set:

$$vol(A) = \sum_{i \in A} d_i$$

Problem: Finding minimum RatioCut and NCut is NP-hard!

Spectral graph partitioning

- Adjacency matrix W :
 - Each entry $w_{ij} = 1$, if (i, j) is an edge, else 0

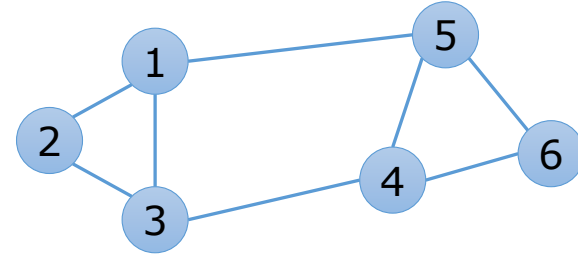


	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	1	1
5	1	0	0	1	0	1
6	0	0	0	1	1	0

symmetric
Diagonal = 0

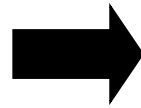
Spectral graph partitioning

- Degree matrix D :
 - Diagonal matrix with $d_{ii} = \sum_j w_{ij}$



W

	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	1	1
5	1	0	0	1	0	1
6	0	0	0	1	1	0



D Degree matrix

	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	3	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

Spectral graph partitioning (two groups, or $k = 2$)

- Remember the goal of RatioCut:

$$\min \text{RatioCut}(A, B) = \min \text{cut}(A, B) \left(\frac{1}{|A|} + \frac{1}{|B|} \right)$$

- Let $\mathbf{y} = [y_1, \dots, y_n]^T$ where $y_i = \begin{cases} +\alpha, & \text{if } v_i \in A \\ -\beta, & \text{if } v_i \in B \end{cases}$

$$\sum_{i=1}^n y_i = 0 \quad \sum_{i=1}^n y_i^2 = n$$

- $\alpha = \sqrt{\frac{|B|}{|A|}}, \quad \beta = \sqrt{\frac{|A|}{|B|}}$ → A组 node数

$$\sum_i y_i = |A| \cdot \alpha + |B| \cdot (-\beta) = |A| \sqrt{\frac{|B|}{|A|}} - |B| \sqrt{\frac{|A|}{|B|}} = \sqrt{|A||B|} - \sqrt{|A||B|} = 0.$$

$$\sum_i y_i^2 = |A| \alpha^2 + |B| \beta^2 = |A| \frac{|B|}{|A|} + |B| \frac{|A|}{|B|} = |B| + |A| = n.$$

Adjacency matrix $\mathbf{W}$:

Each entry $w_{ij} = 1$, if (i, j) is an edge, else 0

Degree matrix $\mathbf{D}$:

Diagonal matrix with $d_{ii} = \sum_j w_{ij}$

Spectral graph partitioning (two groups, or $k = 2$)

- Remember the goal of RatioCut:

$$\min \text{RatioCut}(A, B) = \min \text{cut}(A, B) \left(\frac{1}{|A|} + \frac{1}{|B|} \right)$$

- Let $\mathbf{y} = [\underbrace{y_1, \dots, y_n}]^T$ where $y_i = \begin{cases} \sqrt{\frac{|B|}{|A|}}, & \text{if } v_i \in A \\ -\sqrt{\frac{|A|}{|B|}}, & \text{if } v_i \in B \end{cases}$

- Minimizing the ratio cut is equivalent to optimize the following:

$$\min_{\mathbf{y}} \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}, \mathbf{y}^T \text{ (}\mathbf{y}'\text{'s transposition)}$$

Adjacency matrix $\mathbf{W}$:

Each entry $w_{ij} = 1$, if (i, j) is an edge, else 0

Degree matrix $\mathbf{D}$:

Diagonal matrix with $d_{ii} = \sum_j w_{ij}$

More on theoretical analysis on spectral method:

[Ulrike von Luxburg. A Tutorial on Spectral Clustering](#)



Spectral graph partitioning (two groups, or $k = 2$)

$$\min_{\mathbf{y}} \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}$$

- What is the meaning of $\min \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}$?

- $\mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y} = \sum_{i,j} (d_{ij} - w_{ij}) y_i y_j$
- $= \sum_i d_{ii} y_i^2 - \sum_{(i,j) \in E} 2 y_i y_j$
- $= \sum_{(i,j) \in E} (y_i^2 + y_j^2 - 2 y_i y_j) = \sum_{(i,j) \in E} (y_i - y_j)^2$

$\mathbf{D}$

	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	3	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

By minimizing $\mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}$, we try to make the two ends of an edge assigned to the same label!

Spectral graph partitioning (two groups, or $k = 2$)

- Solving the problem $\min_y \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}$ is hard $O(2^n)$
- We need to enumerate **all possible assignments** of nodes to communities to find the minimum, which is NP-hard
- Approx. the problem to optimize the following (Relaxation):

$$\min_{\mathbf{y} \in \mathbb{R}^n} \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}, \quad s.t. \mathbf{y} \perp \mathbf{1} \text{ (orthogonal)}$$

- $\mathbf{y}$ no longer takes values of either $\sqrt{\frac{|B|}{|A|}}$ or $-\sqrt{\frac{|A|}{|B|}}$, it takes real values!
- This converts the **combinatorial** problem to a **continuous** value optimization problem with some constraint

Recap: eigenvalue and eigenvectors

- Given an $n \times n$ matrix M , a n -dimension column vector x and a real value number λ . λ and x are the eigenvalue and eigenvector of M if

$$Mx = \lambda x$$

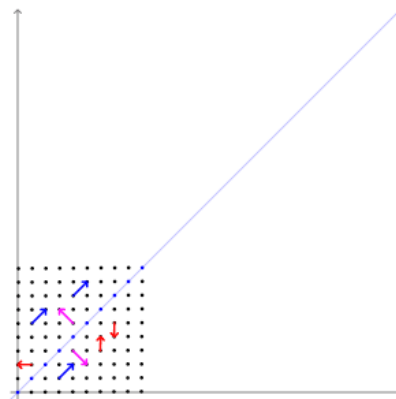
- If M is **symmetric**, λ and x have the following properties:
 - There are n eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Any two distinct eigenvectors are **orthogonal**, i.e., $x_i \perp x_j, i \neq j$

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\det(M - \lambda I) = 0 \quad \lambda_1 = 1, \lambda_2 = 3.$$

$$x_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad x_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$x_1^\top x_2 = \frac{1}{2}(1 \cdot 1 + (-1) \cdot 1) = 0$$



Laplacian matrix

- The graph partitioning objective

$$\min_{\mathbf{y} \in \mathbb{R}^n} \mathbf{y}^T (\mathbf{D} - \mathbf{W}) \mathbf{y}, \quad \text{s.t. } \mathbf{y} \perp \mathbf{1}$$

- Let $\mathbf{L} = \mathbf{D} - \mathbf{W}$. $\mathbf{L}$ is called Laplacian matrix

$\mathbf{W}$

	1	2	3	4	5	6
1	0	1	1	0	1	0
2	1	0	1	0	0	0
3	1	1	0	1	0	0
4	0	0	1	0	1	1
5	1	0	0	1	0	1
6	0	0	0	1	1	0

$\mathbf{D}$

	1	2	3	4	5	6
1	3	0	0	0	0	0
2	0	2	0	0	0	0
3	0	0	3	0	0	0
4	0	0	0	3	0	0
5	0	0	0	0	3	0
6	0	0	0	0	0	2

$\mathbf{L}$

	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix

- The graph partitioning objective

$$\min_{\mathbf{y} \in \mathbb{R}^n} \mathbf{y}^T \mathbf{L} \mathbf{y}, \quad \text{s.t. } \mathbf{y} \perp \mathbf{1} \quad (y_1, y_2, \dots, y_n) \cdot \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = 0$$

- Properties of $\mathbf{L}$ of undirected graph:

- Symmetric**
- The smallest eigenvalue $\lambda_1 = 0$, with corresponding eigenvector $\mathbf{1}$
- For other eigenvectors $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{x} \perp \mathbf{1}$, i.e., $\sum_{i=1}^n x_i = 0$

$\mathbf{L}$

	1	2	3	4	5	6
1	3	-1	-1	0	-1	0
2	-1	2	-1	0	0	0
3	-1	-1	3	-1	0	0
4	0	0	-1	3	-1	-1
5	-1	0	0	-1	3	-1
6	0	0	0	-1	-1	2

Laplacian matrix

- Graph partitioning objective with Laplacian

$$\min_{\mathbf{y} \in \mathbb{R}^n} \mathbf{y}^T \mathbf{L} \mathbf{y}, \quad \text{s.t. } \mathbf{y} \perp \mathbf{1}$$

The minimum value is given by the 2nd smallest eigenvalue λ_2 of matrix $\mathbf{L}$, with $\mathbf{y}$ be the corresponding eigenvector

Justification:

- $\mathbf{y}$ has to be an eigenvector of $\mathbf{L}$, and $\mathbf{y}$ is not $\mathbf{1}$
- If $\mathbf{y}$ is an eigenvector of $\mathbf{L}$: $\mathbf{y}^T \mathbf{L} \mathbf{y} = \mathbf{y}^T \lambda \mathbf{y} = \lambda \mathbf{y}^T \mathbf{y} = \lambda$
- All eigenvectors correspond to their respective eigenvalue
- The 2nd smallest eigenvalue is λ_2

Spectral clustering (bi-partitioning)

- Compute Laplacian matrix for the graph
 - $L = D - W$
- Perform eigenvalue decomposition on L and get the eigenvector y_2 corresponds to the **2nd smallest eigenvalue**.
- Use each dimension in y_2 denotes the label of a node
- Assign nodes with positive values in y_2 to A and nodes with negative values to B.

Example: Spectral Partitioning

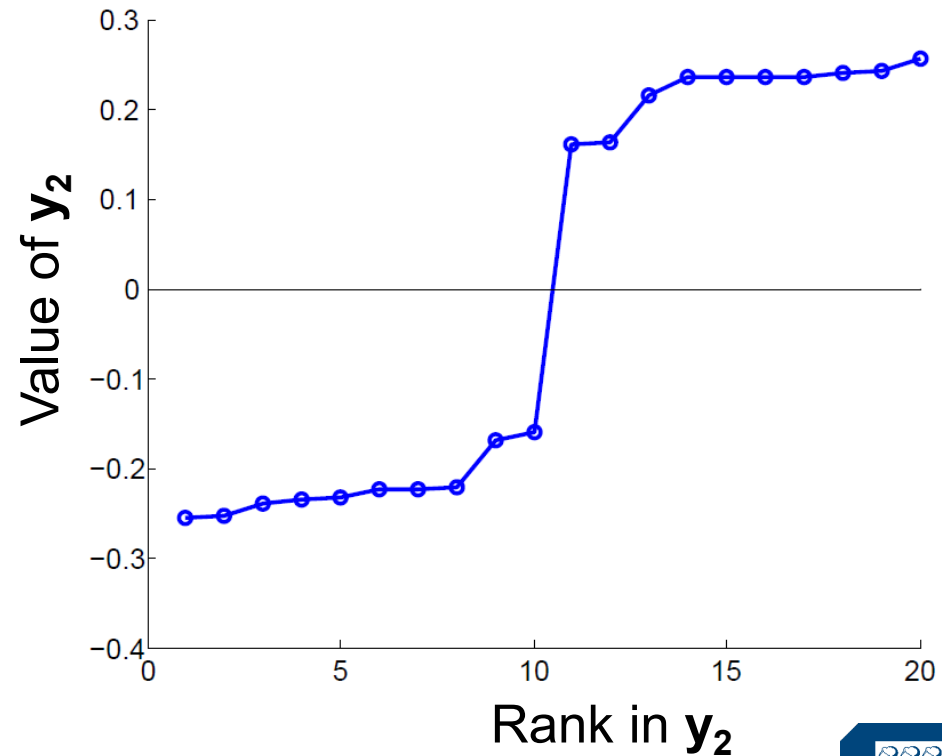
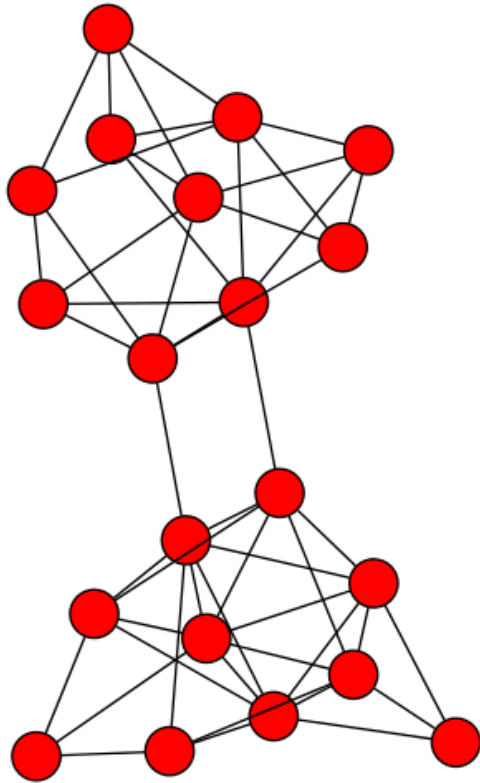
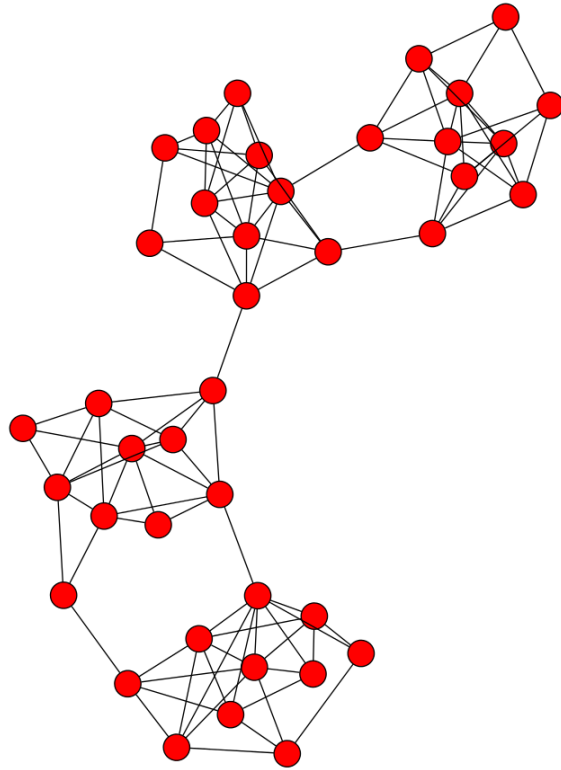


Fig from <http://www.mmds.org/>

Example: Spectral Partitioning



Components of y_2

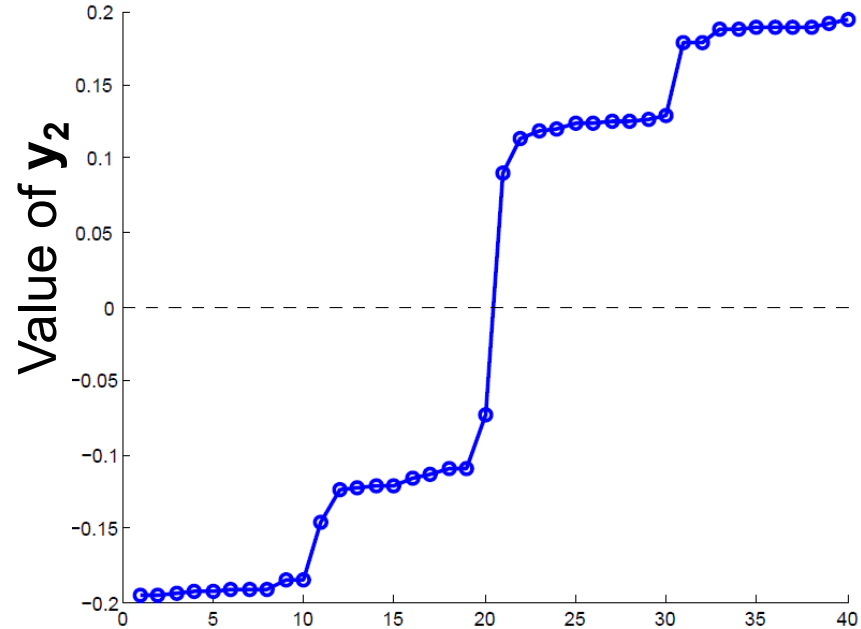
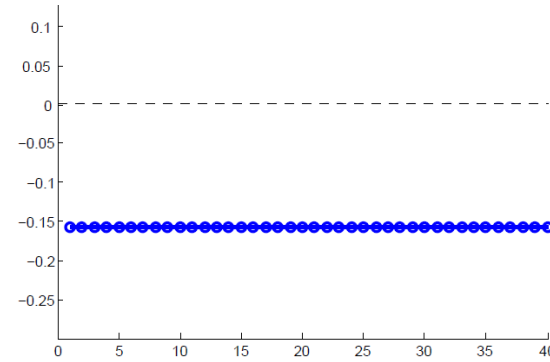
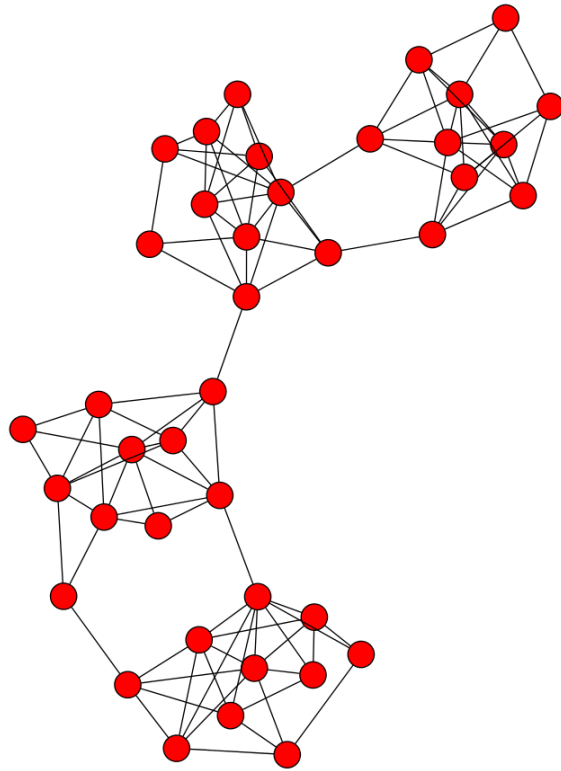


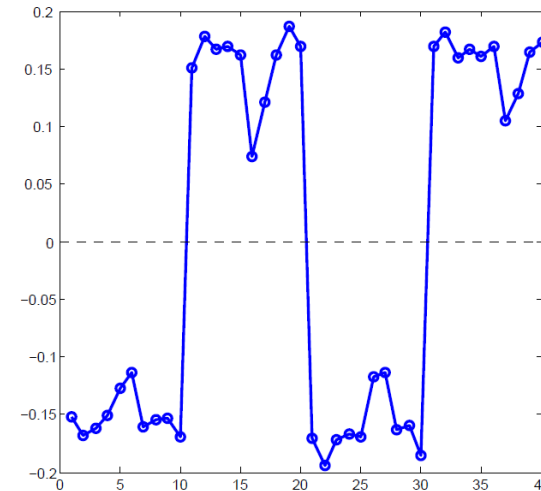
Fig from <http://www.mmds.org/>

Rank in y_2
19

Example: Spectral Partitioning



Components of y_1



Components of y_3

Fig from <http://www.mmds.org/>

Spectral clustering (extension to k-way)

- Compute
 - $L = D - W$ $y_2 \rightarrow \begin{matrix} y_{12} \\ y_{22} \\ y_{32} \\ \vdots \\ y_{n2} \end{matrix}$
- Perform eigenvalue decomposition on L , construct a matrix
 - $Y = [y_2, y_3, \dots, y_k]$, each column y_i is the eigenvector of the i -th smallest eigenvalue
 - Y has n rows, each corresponds to a node $Y \in \mathbb{R}^{n \times k}$
- Use each row to represent a node
 - $v_i = Y[i, :]$
- Apply clustering algorithms (e.g., k-means) on $v_1, \dots, v_n$

Example ($k = 2$)

- Data: $x_1, \dots, x_4$, the corresponding graph is G
 - Cluster into 2 groups

$$W = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 & x_4 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} & \begin{bmatrix} 0 & 5 & 2 & 0 \\ 5 & 0 & 0 & 1 \\ 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix} \end{matrix}$$

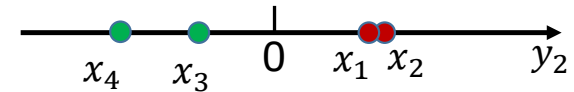
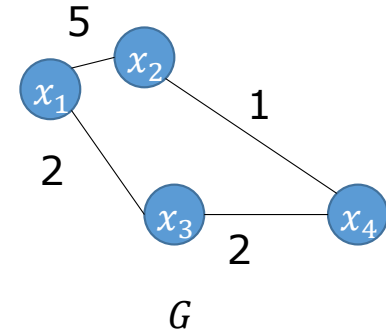
$$D = \begin{bmatrix} 7 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$L = D - W = \begin{bmatrix} 7 & -5 & -2 & 0 \\ -5 & 6 & 0 & -1 \\ -2 & 0 & 4 & -2 \\ 0 & -1 & -2 & 3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 0.705 \\ 0.789 \\ -0.493 \\ -1 \end{bmatrix}$$



$$\begin{aligned} x_1 &= [0.705] \\ x_2 &= [0.789] \\ x_3 &= [-0.493] \\ x_4 &= [-1] \end{aligned}$$



Example ($k = 3$)

- Data: $x_1, \dots, x_4$, the corresponding graph is G
 - Cluster into 3 groups

$$W = \begin{bmatrix} 0 & 5 & 2 & 0 \\ 5 & 0 & 0 & 1 \\ 2 & 0 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

$$L = D - W = \begin{bmatrix} 7 & -5 & -2 & 0 \\ -5 & 6 & 0 & -1 \\ -2 & 0 & 4 & -2 \\ 0 & -1 & -2 & 3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 0.705 \\ 0.789 \\ -0.493 \\ -1 \end{bmatrix} \quad y_3 = \begin{bmatrix} -0.125 \\ 0.512 \\ -1.387 \\ 1 \end{bmatrix} \quad \longrightarrow \quad \begin{aligned} x_1 &= [0.705, -0.125]^T \\ x_2 &= [0.789, 0.512]^T \\ x_3 &= [-0.493, -1.387]^T \\ x_4 &= [-1, 1]^T \end{aligned}$$

